Unit 8 End-of-Unit Assessment - Form A Teacher Key

8 items · 3 points each · 24 total. Award 1 for a valid model/setup, 1 for correct mathematics/units, and 1 for a requested justification. Use the applicable evidence when the task is conceptual; accept equivalent methods and response modes. Do not count writing length as mathematical mastery.

Item 1

Prompt: Test 4 and 5 in $x+3=8$. Which is a solution? Show both statements.

Solution / evidence: $4+3=7$ fails; $5+3=8$ succeeds.

Item 2

Prompt: A collection starts with 20 items and 8 remain. Define used items u, model, solve, and check.

Solution / evidence: $20-u=8$; $u=12$.

Item 3

Prompt: Solve $x+4=16$; justify the same operation on both sides.

Solution / evidence: $x=12$; subtract 4 from both sides and check.

Item 4

Prompt: Solve $n-4=8$; check.

Solution / evidence: $n=12$; add 4 to both sides.

Item 5

Prompt: Solve $4g=28$ and interpret g as equal group size.

Solution / evidence: $g=7$; divide both sides by the group count.

Item 6

Prompt: Solve $t/4=9$ and interpret the total.

Solution / evidence: $t=36$; multiply both sides by 4, then divide to check.

Item 7

Prompt: A measure must be less than 4 kg. Write its constraint and test 3, 4, 5.

Solution / evidence: $w<4$; only 3 of those candidates satisfies.

Item 8

Prompt: At least 4 whole tickets are needed. Write the inequality, graph boundary/direction, and give feasible values.

Solution / evidence: $t\geq 4$; closed boundary at 4, ray right; feasible counts 4,5,...; fractional ticket counts are excluded.

Response: group errors by interpretation/model versus calculation versus explanation. Use the corresponding lesson Teaching Support Guide, revise one step, and reassess with its reserved fresh check. Do not reuse the already seen form as independent evidence.